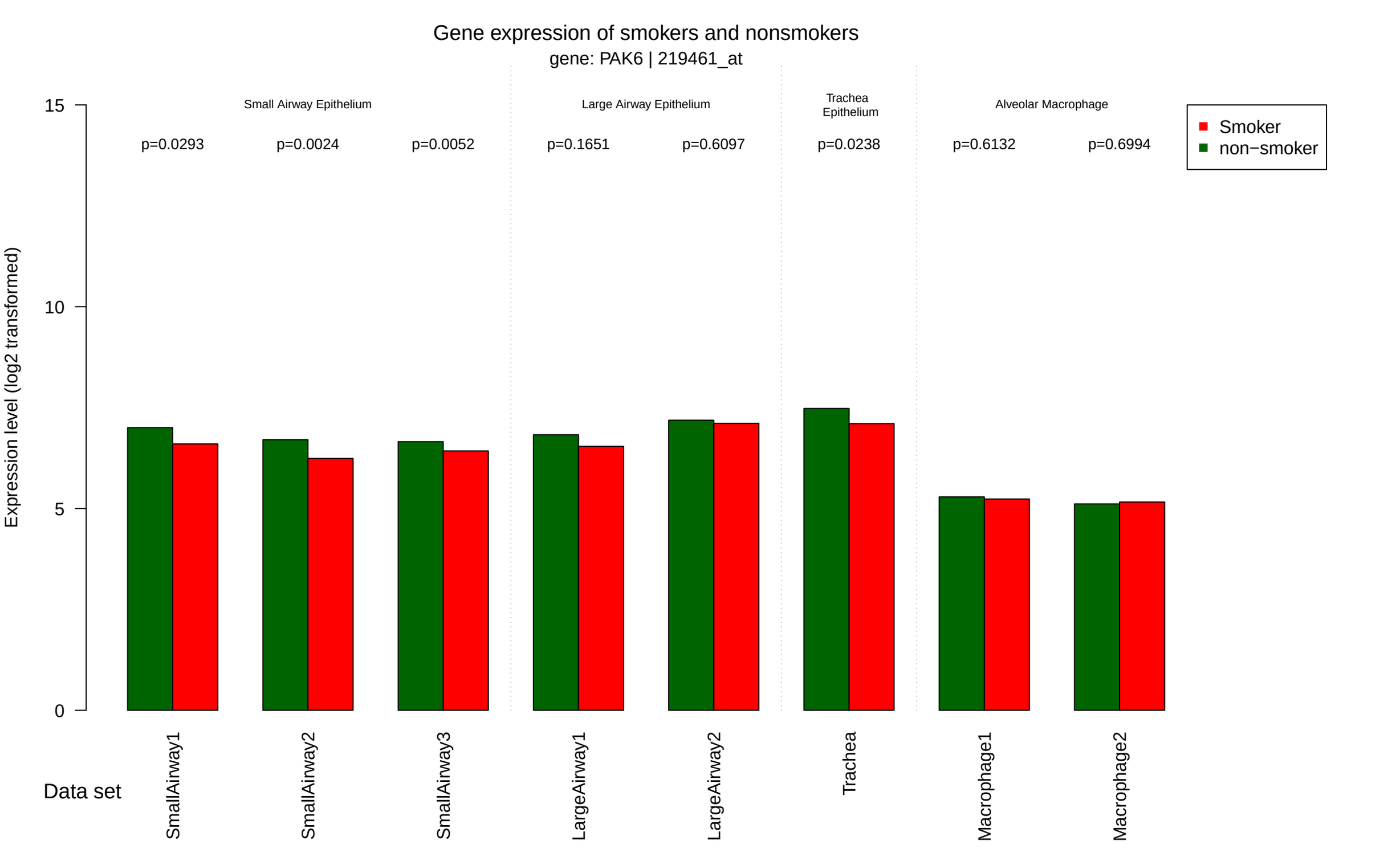
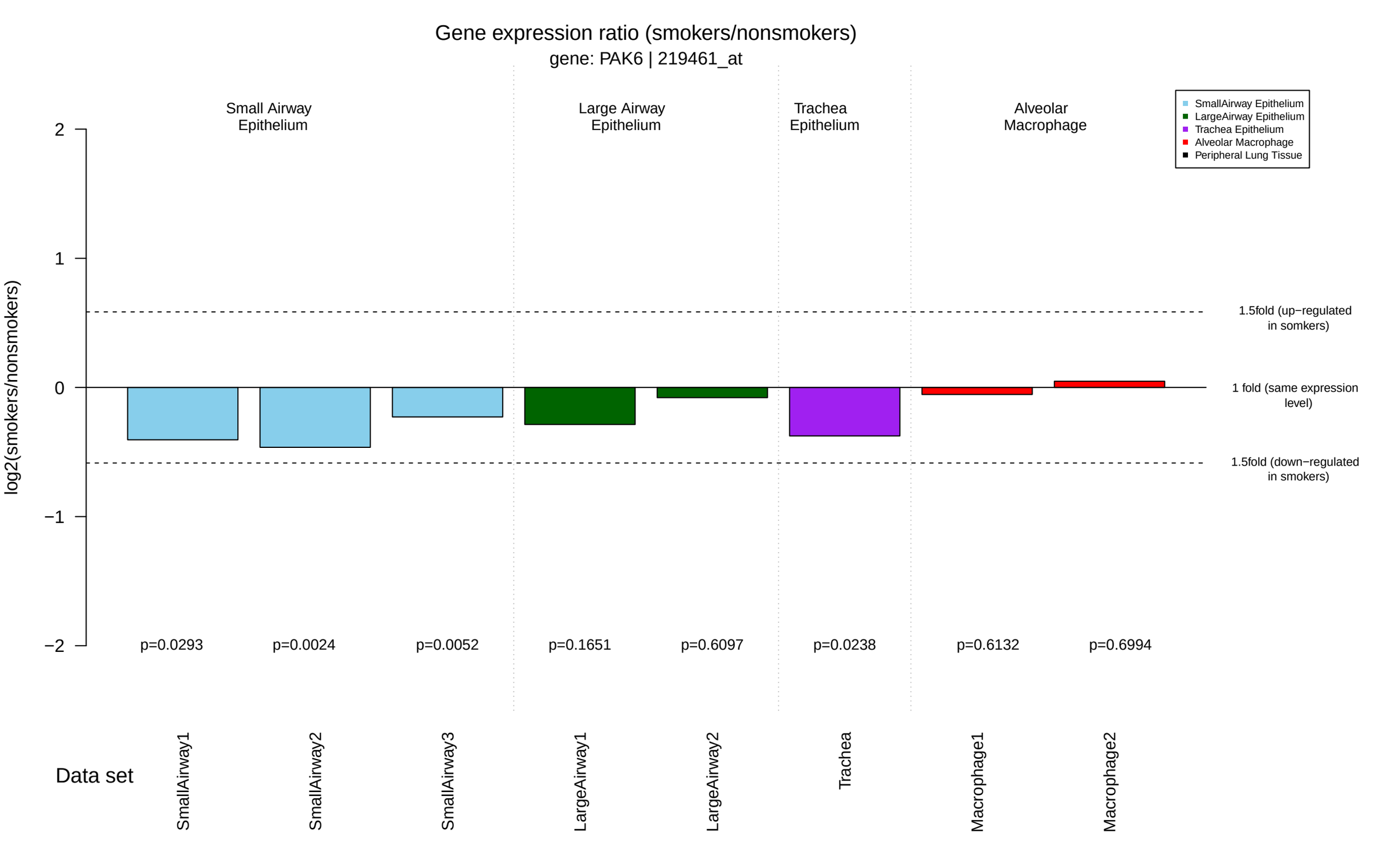
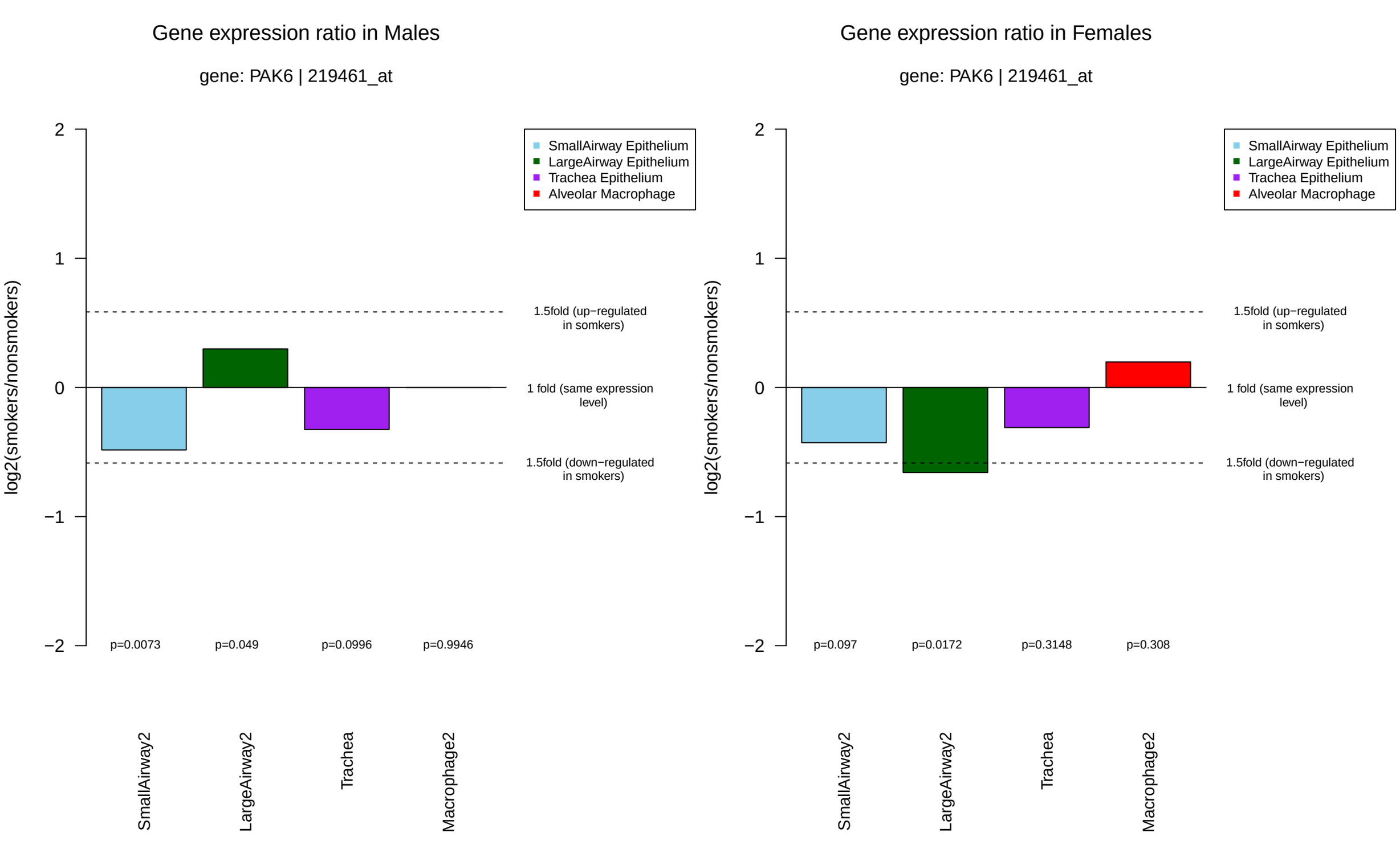
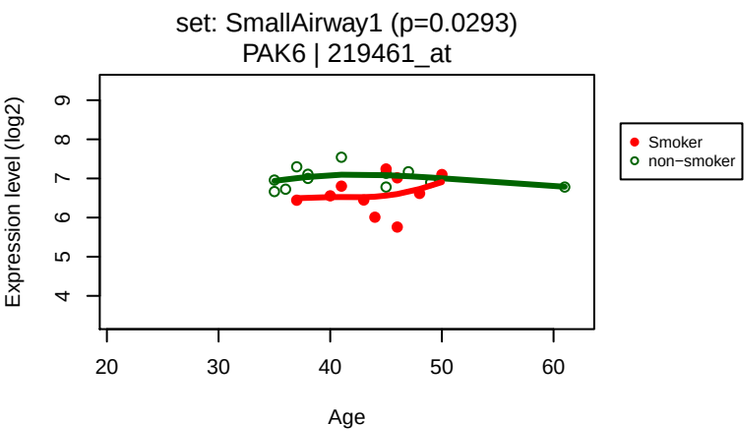
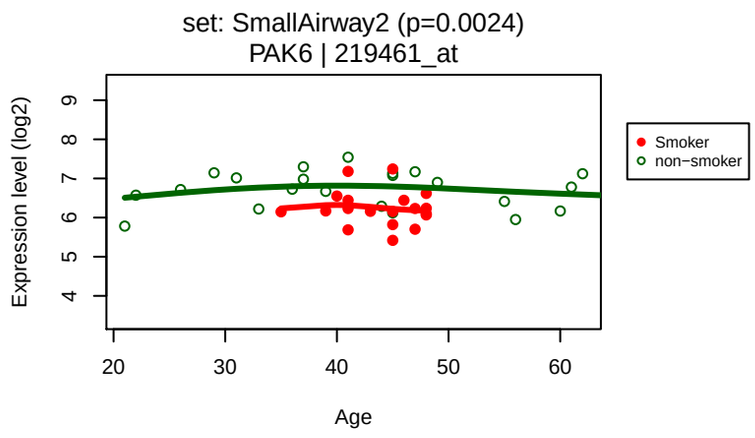
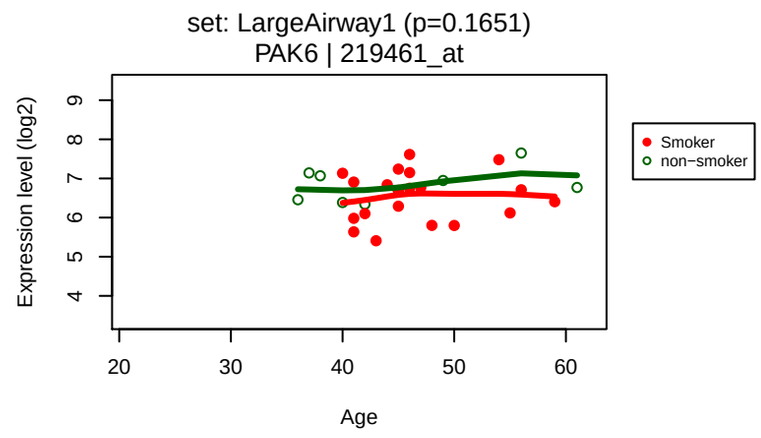

Gene expression of smokers and nonsmokers
gene: PAK6 | 219461_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0293
p=0.0024
p=0.0052
p=0.1651
p=0.6097
p=0.0238
p=0.6132
p=0.6994
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

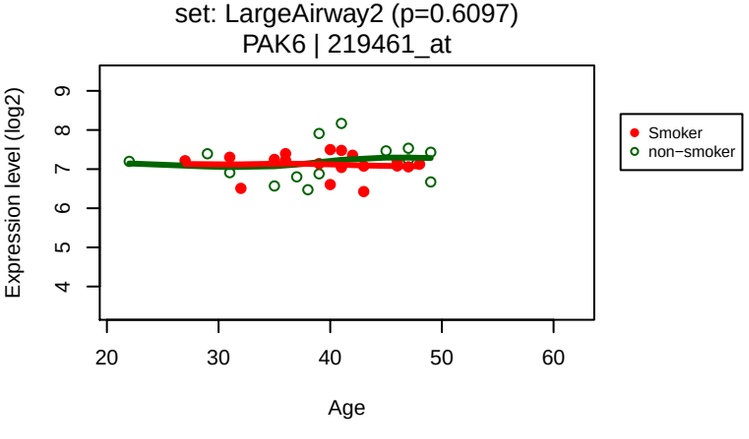

Gene expression ratio (smokers/nonsmokers)
gene: PAK6 | 219461_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0293
p=0.0024
p=0.0052
p=0.1651
p=0.6097
p=0.0238
p=0.6132
p=0.6994
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

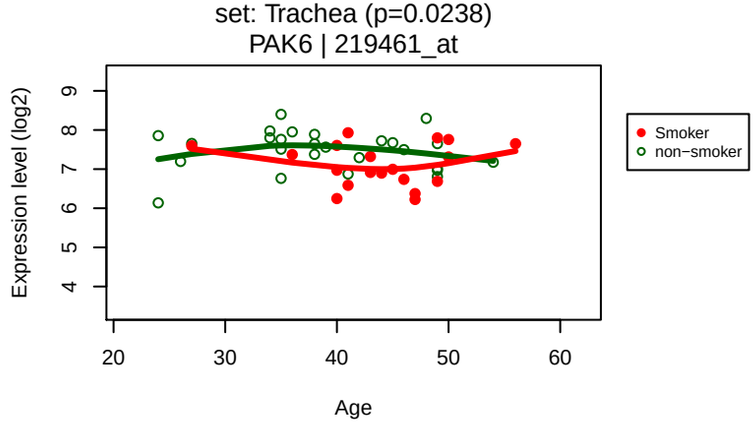

Gene expression ratio in Males
gene: PAK6 | 219461_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0073
p=0.049
p=0.0996
p=0.9946
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAK6 | 219461_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.097
p=0.0172
p=0.3148
p=0.308
SmallAirway2
LargeAirway2
Trachea
Macrophage2

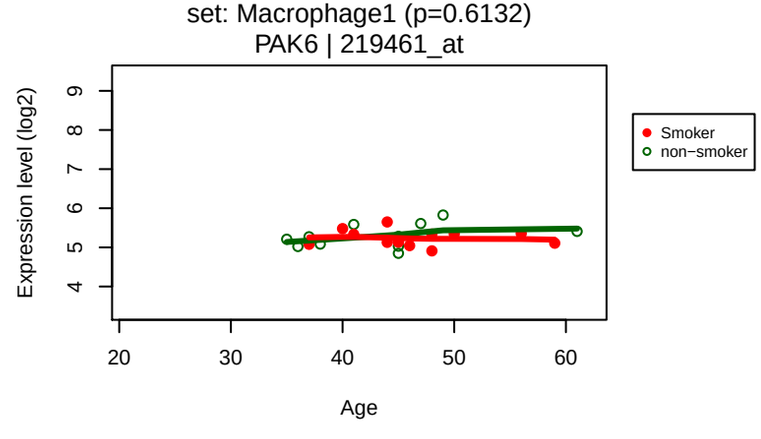

set: SmallAirway1 (p=0.0293)
PAK6 | 219461_at
Expression level (log2)
Age
Smoker
non−smoker

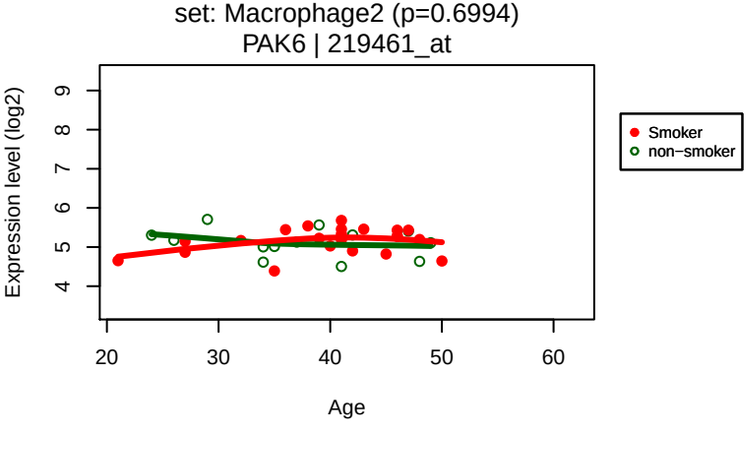

set: SmallAirway2 (p=0.0024)
PAK6 | 219461_at
Expression level (log2)
Age
Smoker
non−smoker

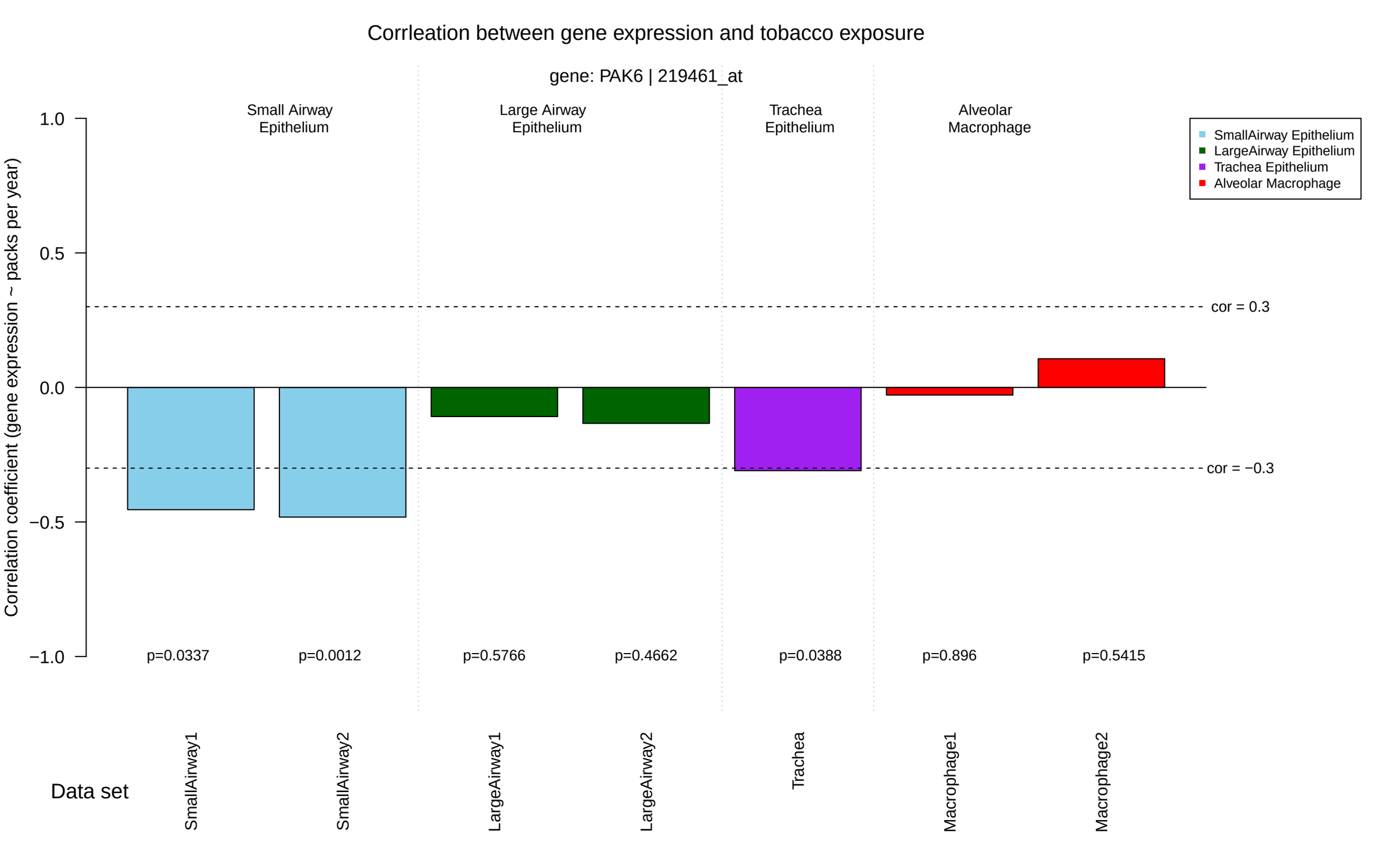

set: LargeAirway1 (p=0.1651)
PAK6 | 219461_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6097)
PAK6 | 219461_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0238)
PAK6 | 219461_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6132)
PAK6 | 219461_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6994)
PAK6 | 219461_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAK6 | 219461_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0337
p=0.0012
p=0.5766
p=0.4662
p=0.0388
p=0.896
p=0.5415
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2